INCH-POUND

MIL-S-20708/1F
13 March 1989

SUPERSEDING
MIL-S-20708/1E
30 July 1987

MILITARY SPECIFICATION SHEET

SYNCHRO, CONTROL TRANSFORMER, TYPE 11CT4E

This specification is approved for use by all Departments and Agencies of the Department of Defense. The requirements for acquiring the Synchros described herein shall consist of this specification and the latest issue of MIL-S-20708.

TABLE I Requirements

Requirement	Value	Unit	Tolerance
Frequency	400	Hz	±1%
Primary Voltage	78 (F)	volts	±1%
Primary Current	18.0	milliamps	maximum
Primary Power	0.24	watts	maximum
Impedance			
Zso	4350-5700	ohms	min-max
Zrs	585-700	ohms	min-max
Impedance Angle			
Zso	77.3-83.5	degrees	min-max
Zrs	22.0-32.0	degrees	min-max
Transformation Ratio	0.735	------	±2%
Phase Shift (Lead)	5.0	degrees	±1.0
Electrical Error	7.0	minutes	maximum
Null Voltage			
Total	60.0	millivolts	maximum
Fundamental	32.0	millivolts	maximum
Friction Torque	0.07	ounce-inches	maximum
Radial Play	0.0006	inches	maximum
End Play	0.0010	inches	±0.0005
Temperature Rise	10.0	degrees (C)	maximum

(F) denotes changes

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TABLE II Military part number output characteristics

Military Part No 1/	A ± 010	B ± 010	G Maximum	Shaft Style	Terminal End	U + 0000 − 0002
M20708/1-01E	555	500	1 732	Splined	Terminal Board	1872
M20708/1-02E	368	312	1 630	Plain & Slotted	Wire Leads	1200
M20708/1-03E	430	.375	↓	↓	↓	↓
M20708/1-04E	493	437				
M20708/1-05E	555	500				
M20708/1-06E	618	562				

(F)

1/Military part number M20708/1-01E shall be the NATO standard

1/The suffix letter following the numerical dash number corresponds to the latest modification letter in the type designation

TABLE III Military part number cross-references

SUPERSEDED MILITARY PART NUMBERS		NEW MILITARY PART NUMBERS
MIL-S-20708/1D	MIL-S-20708/1C	
M20708/1D-001	M20708/1-001	M20708/1-01E
M20708/1D-002		M20708/1-02E
M20708/1D-003		M20708/1-03E
M20708/1D-004		M20708/1-04E
M20708/1D-005		M20708/1-05E
M20708/1D-006		M20708/1-06E

NOTE All line items shown in Table III refer to equivalent and interchangeable synchros of the same type designation modification. Part number changes do not affect form, fit, or function of the synchros listed therein

Custodians:
Army-AR
Navy-AS
Air Force-85

Preparing Activity:
Navy-AS

(Project 5990-0349-1)

Review Activities:
Army-MI,AV
Navy-OS
Air Force-11,17,99
DLA-ES

User Activities:
Navy MC CG

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